

HOST_POWER_MANAGEMENT

Host Power Management — Hard Ban (CONST-033)

Why this exists

On 2026-04-26 18:23:43 the host running mission-critical parallel CLI agents and container workloads was auto-suspended mid-session. This killed the running HelixAgent binary, all 41 dependent services, every SSH session, and every active CLI agent on the box. journalctl showed:

```
systemd-logind[1183]: The system will suspend now!
```

The user-level GNOME power settings were already correct (sleep-inactive-ac-type=nothing). The trigger was the **GDM greeter session at the local console**, which has its own power policy and does not count SSH activity. Earlier, on multiple occasions, the `user@1000.service` had been SIGKILLed by systemd because heavy memory pressure prevented gnome-shell from responding to GDM/ Wayland watchdog within TimeoutStopSec — perceived by the user as "the system fully logged me out." Together these two failure modes have caused repeated loss of in-flight agent work.

The rule

No project shipped from this workspace may invoke a host-level power-state transition. Forbidden invocations include — but are not limited to:

Layer	Forbidden invocations
systemd CLI	<code>systemctl suspend</code> , <code>systemctl hibernate</code> , <code>systemctl hybrid-sleep</code> , <code>systemctl suspend-then-hibernate</code> , <code>systemctl poweroff</code> , <code>systemctl halt</code> , <code>systemctl reboot</code> , <code>systemctl kexec</code>
logind CLI	<code>loginctl suspend</code> , <code>loginctl hibernate</code> , <code>loginctl hybrid-sleep</code> , <code>loginctl suspend-then-hibernate</code> , <code>loginctl poweroff</code> , <code>loginctl halt</code> , <code>loginctl reboot</code>
Legacy CLI	<code>pm-suspend</code> , <code>pm-hibernate</code> , <code>pm-suspend-hybrid</code> , <code>shutdown -h/-r/-P/-H/now</code> , <code>bare reboot/poweroff/halt</code>
DBus	<code>org.freedesktop.login1.Manager</code> . {Suspend, Hibernate, HybridSleep, SuspendThenHibernate, PowerOff, Reboot},

Layer	Forbidden invocations
	<code>org.freedesktop.UPower.{Suspend,Hibernate,HybridSleep}</code> (via <code>dbus-send</code> , <code>busctl</code> , or any language binding)
<code>gsettings</code>	<code>gsettings set ... sleep-inactive-{ac,battery}-type</code> to anything other than <code>'nothing'</code> or <code>'blank'</code>

Defence in depth

Three layers, in order of strength:

- 1. Host-level masking (manual prereq, sudo required, run once):** `scripts/host-power-management/install-host-suspend-guard.sh` masks `sleep.target`, `suspend.target`, `hibernate.target`, `hybrid-sleep.target`, writes `/etc/systemd/sleep.conf.d/00-no-suspend.conf` with `AllowSuspend=no`, and writes `/etc/systemd/logind.conf.d/00-no-idle-suspend.conf` with `IdleAction=ignore` and `HandleLidSwitch=ignore`. After this, no user / session / DE / greeter / cron job can suspend the host.
- 2. User-session bootstrap (no sudo):** `scripts/host-power-management/user_session_no_suspend_bootstrap.sh` runs `gsettings`, `xset -dpms`, and (opt-in via `HOST_POWER_MANAGEMENT_SESSION_INHIBIT=1`) `systemd-inhibit` to protect the current GUI/CLI session of the invoking user. Idempotent. Safe to source from `start.sh` / `setup.sh` / `bootstrap.sh`.
- 3. Source-tree static gate:** `scripts/host-power-management/check-no-suspend-calls.sh` walks the tree and exits non-zero on any forbidden invocation. `challenges/scripts/no_suspend_calls_challenge.sh` wraps it as a challenge that runs in CI / `run_all_challenges.sh`. `challenges/scripts/host_no_auto_suspend_challenge.sh` asserts the running host's state matches the layer-1 masking.

Verification

```
# Layer 1: host state
bash challenges/scripts/host_no_auto_suspend_challenge.sh
# Expected: 4 PASS

# Layer 3: source tree
bash challenges/scripts/no_suspend_calls_challenge.sh
# Expected: PASS, no forbidden calls
```

If you genuinely need a power-state transition

You don't, in this workspace. If a future legitimate use case arises (e.g. a container's *internal* init script that suspends a *guest VM*, not the host), add the specific file path to `EXCLUDE_PATHS` at the top of `check-no-suspend-calls.sh` with a comment explaining the non-host context.